INSTITUTE OF DATA · CAPSTONE · APRIL 2026

Stock Gossip Monitor

“If you don’t master AI coders, you’re a fking retard.”**
June 2025, ex Amazon Product Manager

3
Core Pages

9
API Routes

30k+
stocks covered

Industry Backdrop

CONTEXT

Traditional Equity Investors

- 3–5 year investment horizon
- Edge: patient capital + variable sizing
- Outperformance comes in spurts — hard to consistently outperform

Competition

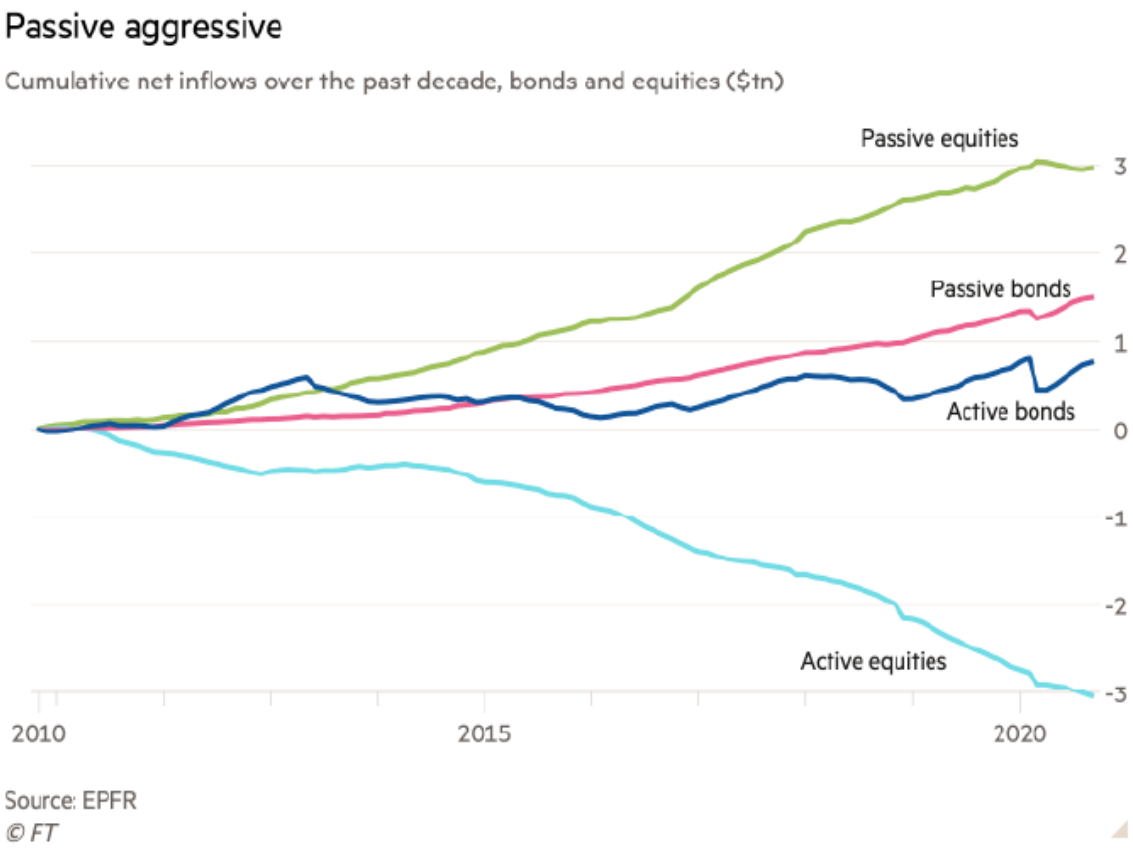
- **Pods:** short-term momentum + tight risk control
- **Quants:** algorithm-driven factor trading
- Both squeeze the traditional investor's edge

The Core Problem

- **Macro** (geopolitics) + **factors** (industry, financial investment categories) **drive most returns, firm-specific a small driver**
- Traditional investors need patient capital to add investment value, but this requires **providing investors with valuable insights**
- Yet **traditional investors waste too much time on commoditized firm-specific research that should be semi-automated**

Industry Backdrop

CONTEXT



Problem & Solution

HYPOTHESIS

The Problem

Incumbent software: **too broad, too much**

Investors **need to focus on metrics tailored to each investment category**

Deep research takes time and effort on Excel – **Excel doesn't scale well**

The Solution

WHO	**1–5 person boutique teams, family offices, self-directed HNW investors**
WHAT	**Harness for category-specific metrics** with **user overrides**
END VISION	Multi-API community research tool drawing from variety of data sources, **accessible anywhere**, AI-agent ready

Free up time for traditional equity investors to think more clearly and make more confident portfolio decisions

Product requirements: high level

DESIGN GOALS

01

Ease of Use

→ No onboarding required

→ Minimalist user interface

02

Accurate Relevant Data

→ **Relevant data** for each stock **must be accurate - irrelevant data** for each stock **will not be displayed**

→ **User overrides for most values, calculations for key values**

03

Heavy Documentation

→ Prepare for future human developer/ AI coder input

→ DB schema, architecture and test/ script summary **markdowns should auto-update on code change**

Simple logic + accurate, customizable data + documentation = ready for future AI coder input and AI agent accessibility

Product requirements: User Flow

USER FLOW

1 Home Page Investment Category Performance Charts

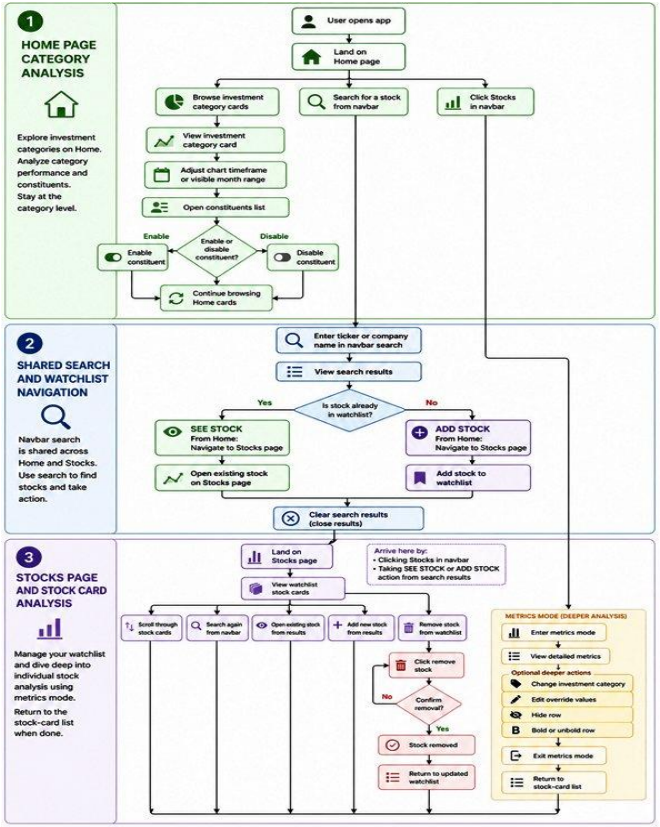
- Browse how different investment categories have performed
- Adjust timeframe of analysis (indexed to 100 from starting point)
- Enable/ disable individual stock constituents

2 Shared Search & Watchlist Nav

- Search by ticker or company name
- SEE STOCK (if in watchlist) or ADD STOCK (to watchlist)
- Accessible from Home + Stocks pages

3 Stocks Page Deep Card Analysis

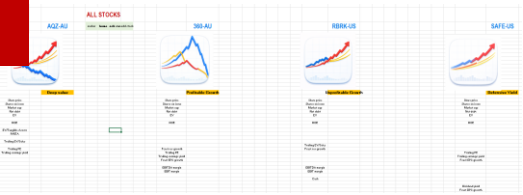
- View and manage your individual stock card (change Inv Categories)
- Customize what metrics you want to view in the future (override values, bold/ unbold rows, show/ hide rows)



Wireframes & prototypes

STEP BY STEP

0



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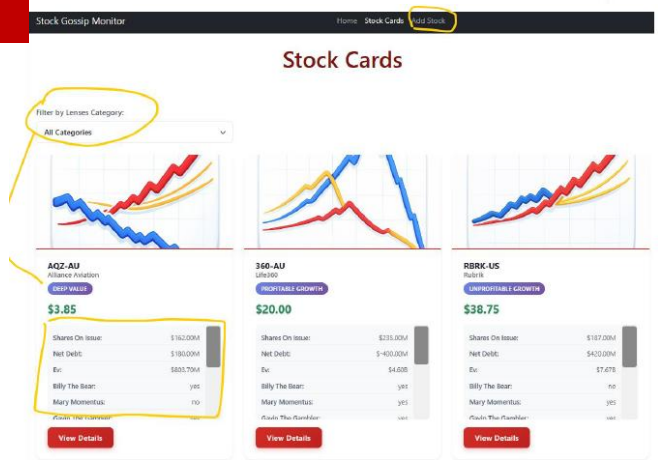

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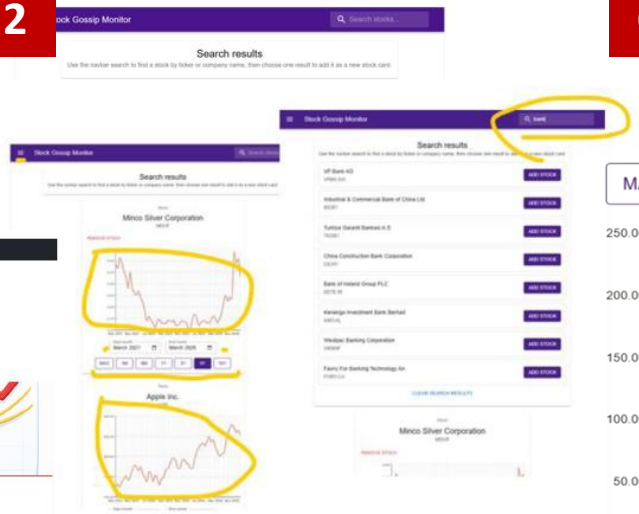

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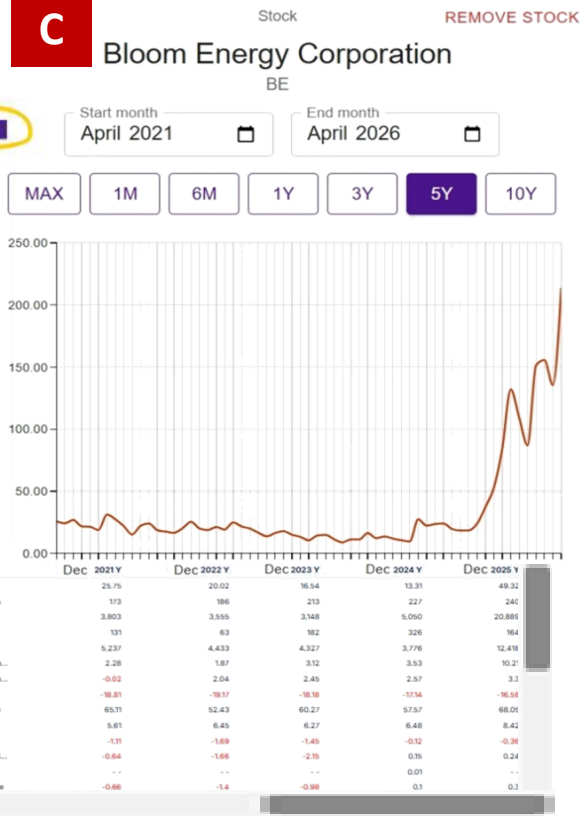
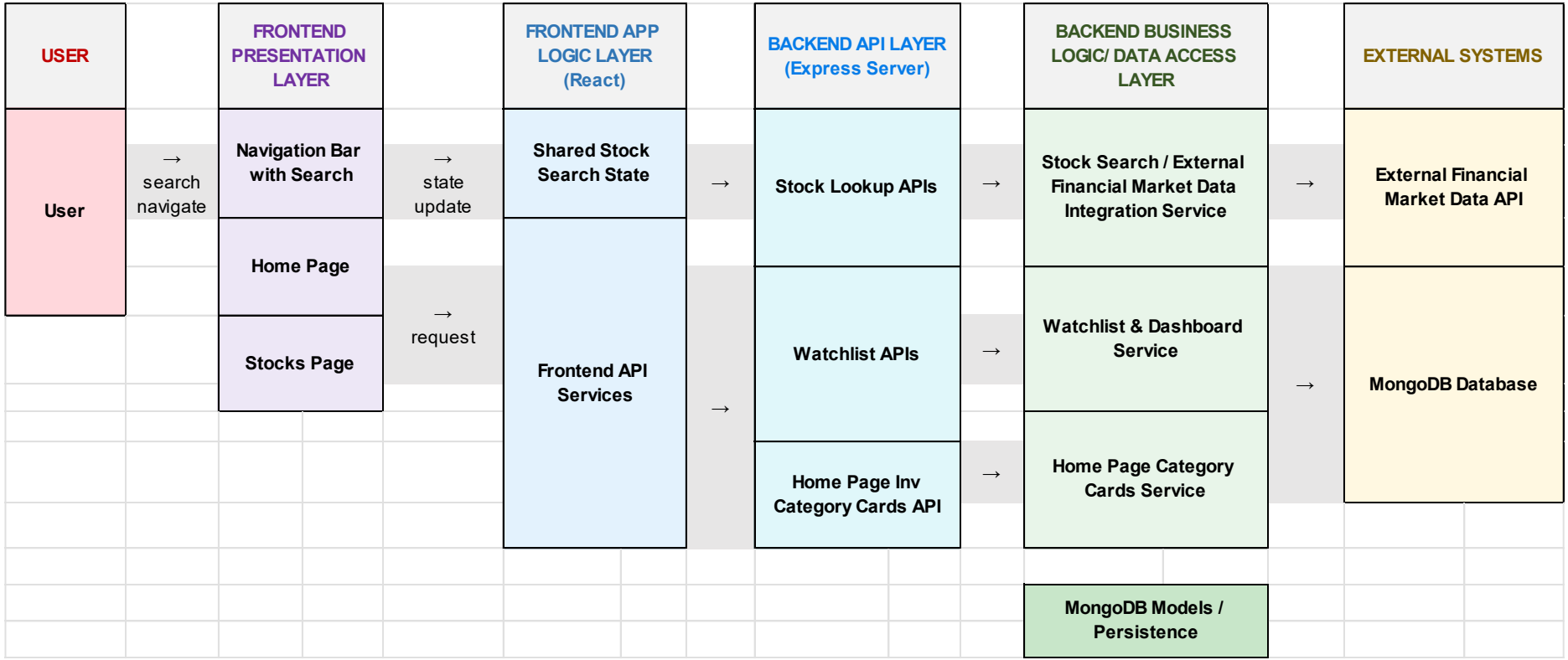


In an AI world, interactive working prototypes, not wireframes are the fastest way to securing stakeholder buy-in.

Architecture

DIAGRAM



Testing, Quality & Doc Maintenance Safety Nets

QA

✅ Automated Tests

→ **Backend: Node built-in test runner** instead of manual Thunder Client

API route coverage (all CRUD)

Investment category lens validation

→ **Frontend: Vitest test runner** + Testing Library on JSDOM fake browser

🔬 Diagnostic Scripts

→ **Live CLI search harness** (external financial market data API) **to finetune search algo**

→ **Custom backend memory benchmark** (2,000 stocks)

→ **Playwright real browser frontend test** (2,000 stocks)

📄 Documentation

→ Schema ref doc, architecture doc, test & script **markdown documents auto-regenerate on app change**

→ Onboarding, maintenance and collaboration with **human developers and AI coders**

→ Critical for **multi-API/ agent harness aspirations**

Demo

Q&A

Tech Stack & References

STACK

External Financial Market Data API

Primary data source

MongoDB

Persistent app data store

Mongoose

Schema modeling + rules

React

Component UI + shared state

Express

API server + routing

Vitest

Frontend unit tests

Testing Library

Component interaction tests

Node test runner

Backend route tests

Playwright

E2E + browser perf benchmark

Vite

Build tooling + dev server

Key Backend Items

BUSINESS LOGIC | DATA ACCESS

External Financial Market Data Integration Service

- Ticker vs company name search logic
- Handles API errors + data reshaping
- CLI harness for isolated testing

Watchlist & Dashboard Service

- Load + build stock card data
- Apply saved row preferences
- Refresh data on demand

Category Cards Service

- Equal-weighted constituent performance
- Enable / disable individual stocks
- Extensible: War/Peace, Tech/Commodities…

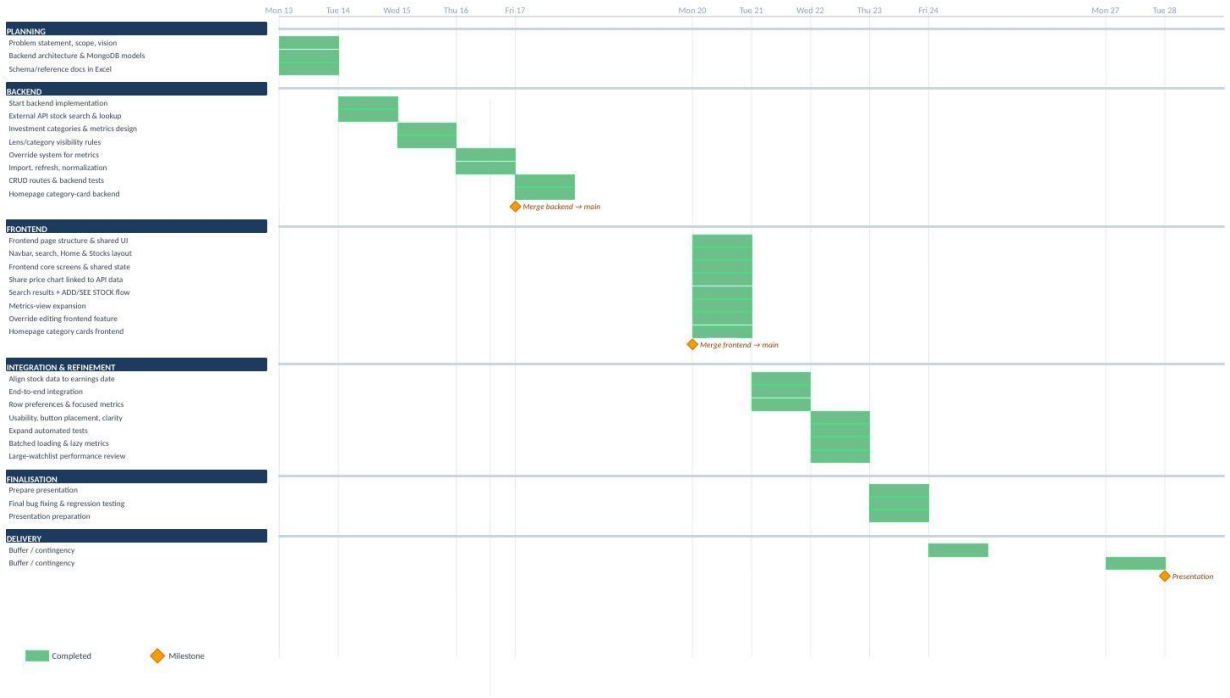
MongoDB Models / Persistence

- Mongoose schemas define data shape
- Handles multi-API data conflicts
- Schema docs auto-update on every change for human/ AI developer ease-of-use

Project Planning

“Everyone has a plan until they get punched in the face”

TIMELINE



🐛 Buffer → Bugs

All contingency time consumed on backend integration, infinite React loops + performance scaling issues

📈 Scale achieved

2,000 stocks handled in both frontend and backend

💡 Key lesson

Write tests alongside code. **Automate away manually discovered errors.** Never let errors compound.